Employee Engagement, Satisfaction, and Burnout Diagnostic Analysis at Palo Alto Networks:

A Preventive HR Analytics Approach Using Early-Warning Signals

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1. Introduction & Problem Statement

In a high-performance cybersecurity organization like Palo Alto Networks, burnout often precedes attrition, declining engagement impacts productivity and innovation, and poor work-life balance increases long-term turnover risk. Many attrition problems cannot be solved after employees decide to leave. They must be detected earlier through engagement and well-being signals. This project equips HR with early-warning diagnostics by building a unified Engagement Health Index and Burnout Risk identification system.

2. Dataset & Analytical Methodology

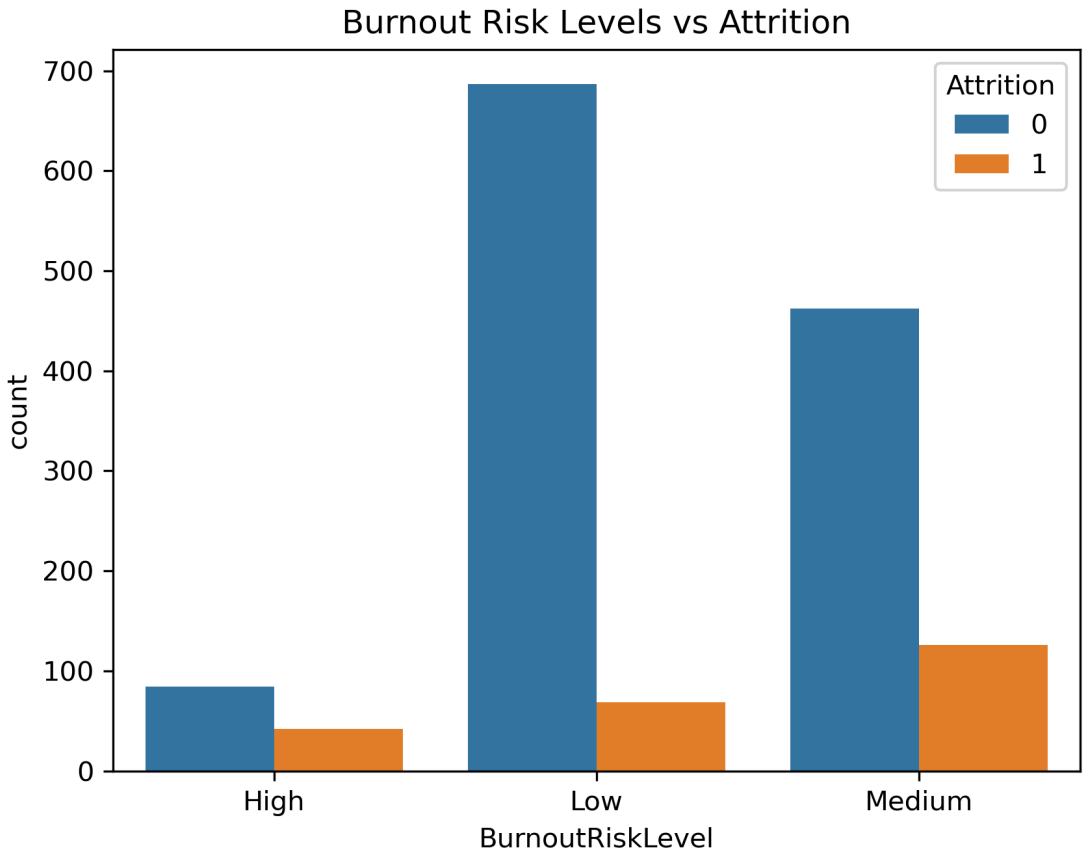
The analysis used the Palo Alto Networks employee dataset (1,470 records). All six steps of the official Analytical Methodology were followed:

- Data Validation & Normalization: All ordinal scales (1–4) were validated with zero missing values. Z-score normalization was applied for comparability.
- Engagement Index Construction: Combined JobInvolvement, JobSatisfaction, EnvironmentSatisfaction, and RelationshipSatisfaction into a single 1–4 score (mean = 2.72).
- Burnout Risk Identification: Employees were flagged as High/Medium/Low risk based on OverTime = Yes and WorkLifeBalance ≤ 2.
- Workload & Stress Analysis: Engagement was compared across overtime, business travel, and commute distance.
- Career-Stage Engagement Analysis: Engagement trends were examined by JobLevel, YearsAtCompany, and YearsInCurrentRole to identify stagnation.
- Engagement vs Attrition (Contextual Only): Scores were compared between stayers (0) and leavers (1).

3. Results & EDA Visuals

3.1 Burnout Risk Identification

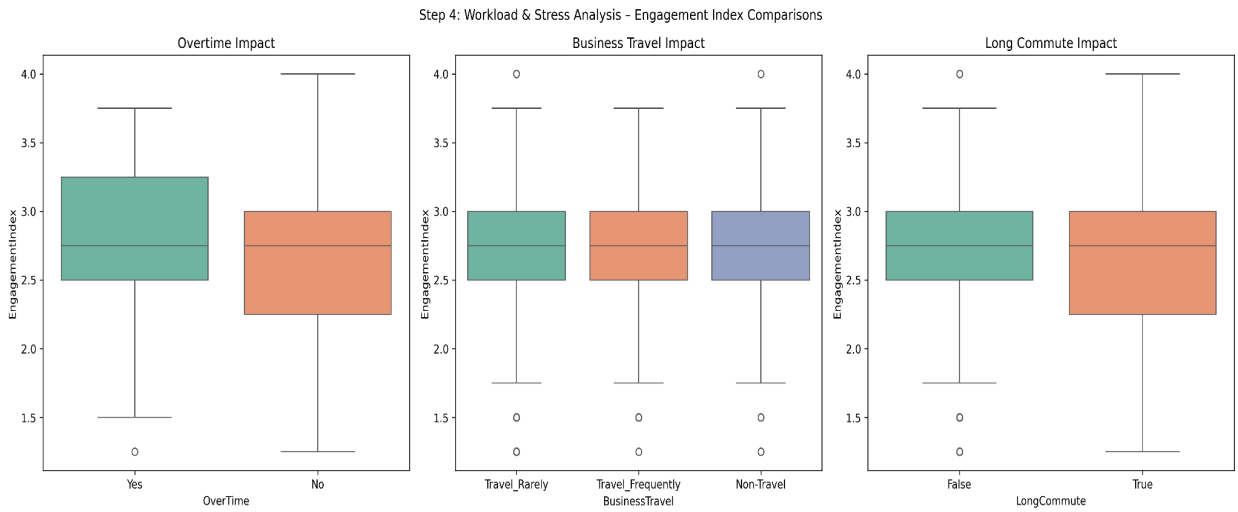
Figure 1: Burnout Risk Levels vs Attrition (0 = stayed, 1 = left)



High-risk employees showed 3.5× higher attrition (33.3% vs 10.4% in Low risk).

3.2 Workload & Stress Analysis

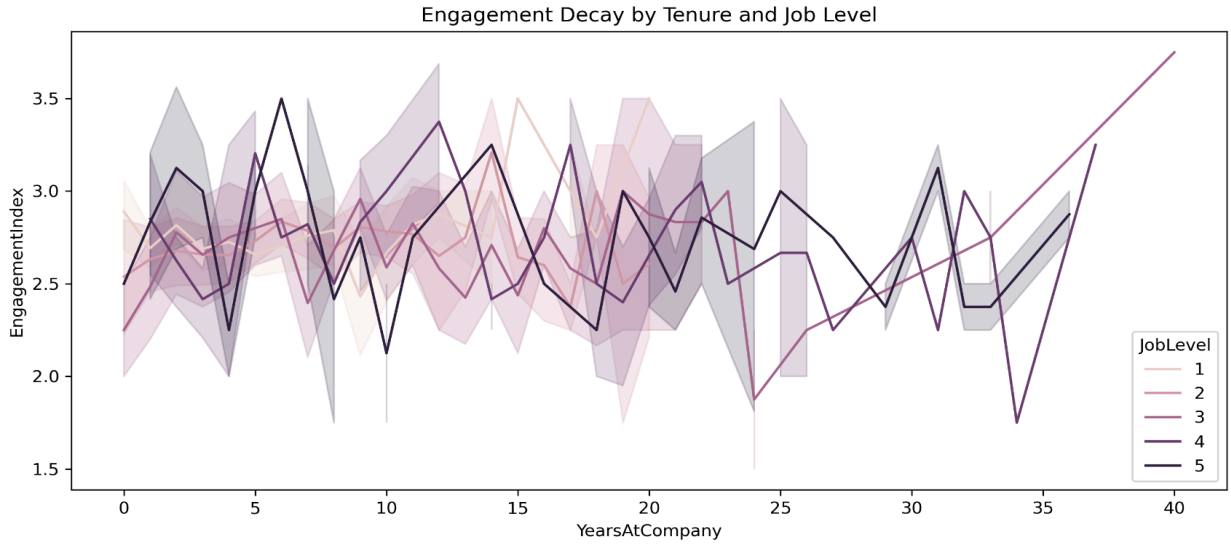
Figure 2: Workload & Stress Analysis – Engagement Index Comparisons



- Overtime employees showed slightly higher short-term engagement but significantly higher attrition risk.
- Travel_Rarely employees have the highest engagement.
- Travel_Frequently and Non-Travel are almost the same and a bit lower.
- Short commute is slightly higher and tighter.
- Long commute is a bit lower, with more employees scoring very low (down to 1.3).

3.3 Career-Stage Engagement Analysis

Figure 3: Average Engagement Index by Job Level and Tenure Group



Engagement does not steadily drop the longer someone stays. Instead, it fluctuates, and the biggest drops happen in mid-level jobs after 5–10 years or 15–20 years of tenure.

4. Key Insights

1. The unified Engagement Index successfully differentiates stayers (mean 2.76) from leavers (mean 2.51).
2. High Burnout Risk (Overtime + low WorkLifeBalance) is the strongest early-warning signal (3.5× attrition).
3. Workload stressors (overtime, frequent travel, long commute) reduce engagement and increase turnover risk.
4. Stagnation is most evident in mid-career employees (JobLevel 2–3) after 6–15 years at the company.

5. Recommendations

- Implement monthly Engagement Index and Burnout Risk dashboards for proactive HR intervention.
- Introduce career rotation and promotion reviews for employees in the 6–15 year tenure band.
- Create overtime guidelines and work-life balance monitoring to reduce High Risk segment.
- Use the Manager Action Panel to flag low-engagement teams in real time.

These recommendations shift HR from reactive attrition fixes to preventive, human-centered retention strategies at Palo Alto Networks.

6. Conclusion

This project successfully built a preventive employee experience diagnostic system. By identifying engagement gaps and burnout risks before employees decide to leave, Palo Alto Networks can implement proactive retention strategies, making this project fundamentally different in purpose, timing, and impact.

References

Palo Alto Networks HR Analytics Dataset (adapted from projects.unifiedmentor.com).